MATH 10A — STRUCTURE AND CONCEPTS IN MATH I

§6.4

Multiplying and Dividing Fractions

The equal groups model, the area model, quotative and partitive division, and Keep-Change-Flip

Today's Goal

After completing this section, you should be able to:

- Multiply a whole number by a fraction and a fraction by a whole number using the equal groups model
- Multiply a fraction by a fraction using the area model
- Divide with fractions using quotative and partitive division with the bar model and the area model
- Divide fractions using the Keep-Change-Flip strategy and by dividing numerators and dividing denominators

Learning Target 15

LT.15 Arithmetic with Fractions

I can apply standard strategies for addition, subtraction, multiplication, and division of fractions and rational numbers using appropriate representations.

Using the Equal Groups Model

Do $2 \times \frac{3}{7}$ and $\frac{3}{7} \times 2$ mean the same thing? What does each number tell us?

$2 \times \frac{3}{7}$

When multiplying a fraction by a whole number, we may think of this using the equal groups model with the whole number telling us how many groups.

$\frac{3}{7} \times 2$

When we consider a fraction times a whole number, now the fraction represents the number of groups.

A Whole Number Times a Fraction

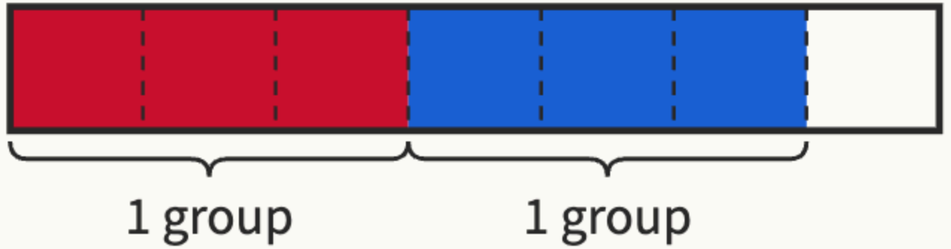
Use the equal groups model to evaluate $2 \times \frac{3}{7}$.

This expression means "2 groups of size $\frac{3}{7}$". So we begin by representing each group using an appropriate model. In our case, we will use the bar model.

We divide the whole bar into seven equal parts.

We shade 1 group of 3 of these parts.

We shade another group of 3 of these parts.



So $2 \times \frac{3}{7} = \frac{6}{7}$.

A Fraction Times a Whole Number

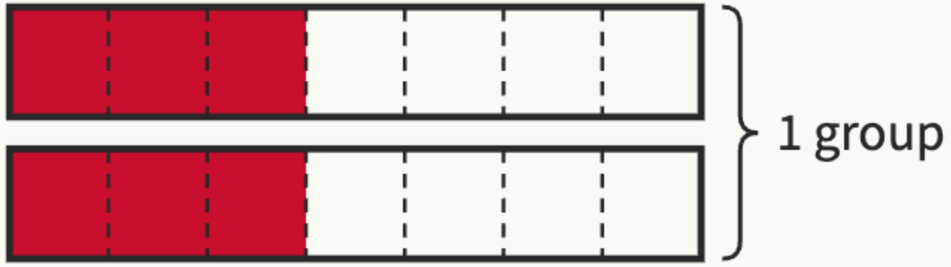
Use the equal groups model to evaluate $\frac{3}{7} \times 2$.

Each group has size 2, but we don't even have one whole group. We only have $\frac{3}{7}$ of a group.

We begin by representing one group using the bar model. Remember that each bar represents one whole, not one group.

Since we want $\frac{3}{7}$ of a group, we will divide each whole into seven equal parts.

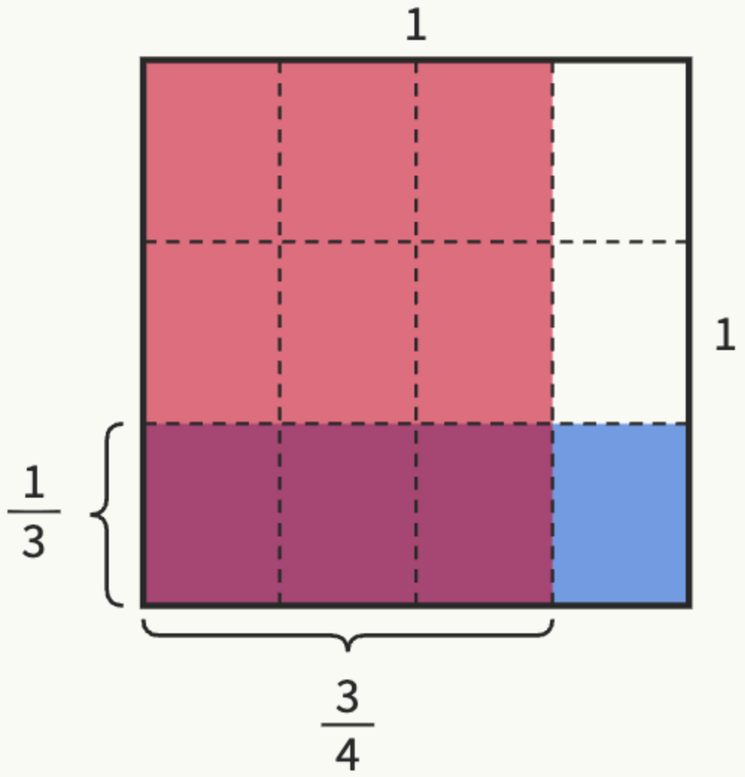
We take $\frac{3}{7}$ of each whole.



So $\frac{3}{7} \times 2 = \frac{6}{7}$.

Using the Area Model

Use the area model to evaluate $\frac{1}{3} \times \frac{3}{4}$.



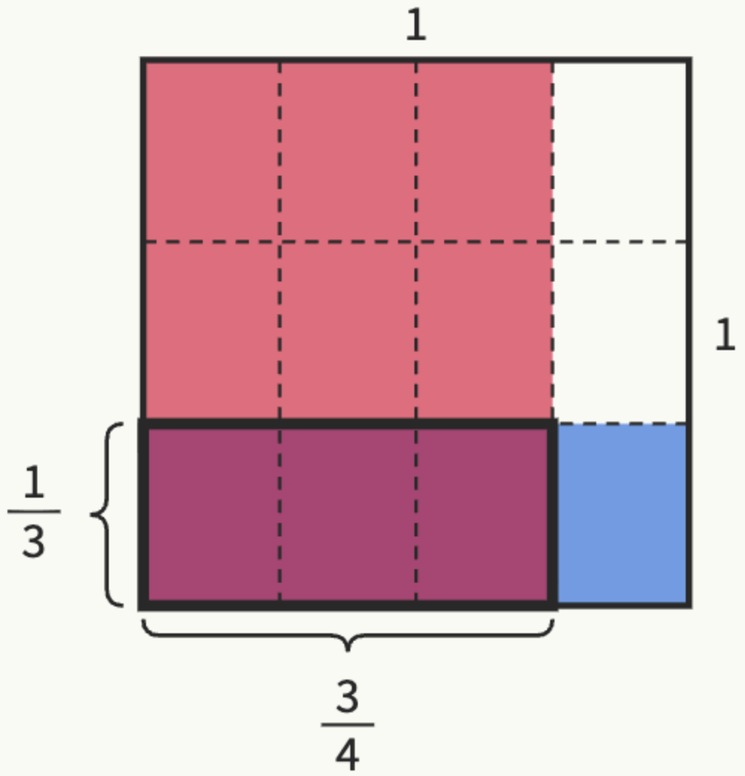
We represent 1×1 by a square whose sides are each 1 unit long.

First we divide one side of the square into 3 equal parts.

We also divide the other side of the square into 4 equal parts.

Using the Area Model

Use the area model to evaluate $\frac{1}{3} \times \frac{3}{4}$.



Notice that the part of the square that is shaded blue and red represents the product $\frac{1}{3} \times \frac{3}{4}$.

Since the area of the square, which is 1, has been divided into 12 equal parts and the rectangle representing $\frac{1}{3} \times \frac{3}{4}$ consists of 3 of these parts, it follows that $\frac{1}{3} \times \frac{3}{4} = \frac{3}{12}$.

Multiplying Two Fractions

How could we get $\frac{1}{3} \times \frac{3}{4} = \frac{3}{12}$ without drawing the square?

It seems like we can just multiply the numerators and multiply the denominators: $\frac{1}{3} \times \frac{3}{4} = \frac{1 \times 3}{3 \times 4} = \frac{3}{12}$. This is in fact true and is justified by the area model.

When we multiply the denominators, that gives us the total number of equal parts the square is divided into.

When we multiply the numerators, that gives us the total number of those equal parts that are needed for the rectangle whose area represents the product of the two fractions.

Quotative and Partitive Division

In $12 \div 4$, what is unknown in the quotative model? What is unknown in the partitive model?

Quotative

In the quotative model for division, the number of groups is unknown. So we interpret $12 \div 4$ as "how many groups of 4 are in 12?" We can use this same interpretation when dividing a whole number by a fraction.

Partitive

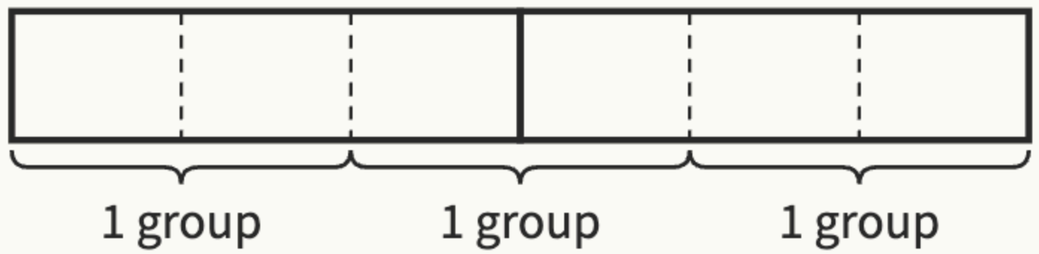
In the partitive model for division, the size of each group is unknown. We can use the partitive model when dividing a fraction by a whole number.

A Whole Number Divided by a Fraction

Use quotative division to evaluate $2 \div \frac{2}{3}$.

We are looking for how many groups of size $\frac{2}{3}$ are in 2. We can represent 2 using the bar model, where each bar represents one whole.

Since we are looking for groups of $\frac{2}{3}$, we divide each whole into 3 equal parts.



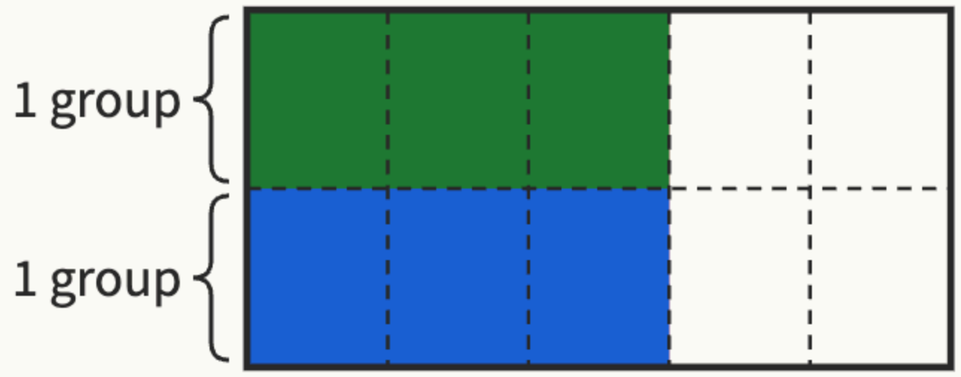
There are 6 parts in all, and each group uses 2 of them. So $2 \div \frac{2}{3} = 3$.

A Fraction Divided by a Whole Number

Use the partitive division model to evaluate $\frac{3}{5} \div 2$.

We are looking for the group size when $\frac{3}{5}$ is divided into 2 equal groups. We can represent $\frac{3}{5}$ using the area model with a rectangle representing one whole.

Since we need to split $\frac{3}{5}$ into two equal groups, we split each of the five parts of our area model into 2 equal parts by drawing a horizontal line through the center of the rectangle.



This creates two equal groups of size $\frac{3}{10}$. So $\frac{3}{5} \div 2 = \frac{3}{10}$.

A Fraction Divided by a Fraction

Use the quotative division model to evaluate $\frac{5}{7} \div \frac{3}{7}$.

When dividing a fraction by another fraction, it helps to think of the unit fractions involved.

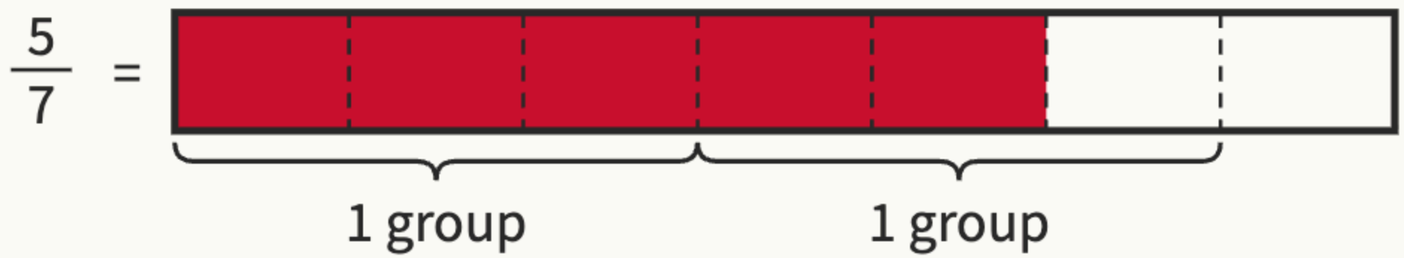
Note that both fractions are in terms of the same unit fraction: one-seventh. This makes things much easier for us.

We begin by representing $\frac{5}{7}$ using the bar model.

$\frac{5}{7} =$ 

A Fraction Divided by a Fraction

Use the quotative division model to evaluate $\frac{5}{7} \div \frac{3}{7}$.



Then we look for groups of size $\frac{3}{7}$. We can see one complete group in the shaded region.

However, there is not enough in the shaded region for another complete group. For the second group, the shaded region only gives us $\frac{2}{3}$ of what is needed.

So we have a total of $1\frac{2}{3}$ groups, which means that $\frac{5}{7} \div \frac{3}{7} = 1\frac{2}{3}$.

Different Denominators

Use the quotative division model to evaluate $\frac{2}{3} \div \frac{2}{5}$.

We will rewrite each fraction to have the least common denominator, which is 15.

$$\frac{2}{3} = \frac{2 \cdot 5}{3 \cdot 5} = \frac{10}{15} \qquad \text{and} \qquad \frac{2}{5} = \frac{2 \cdot 3}{5 \cdot 3} = \frac{6}{15}$$

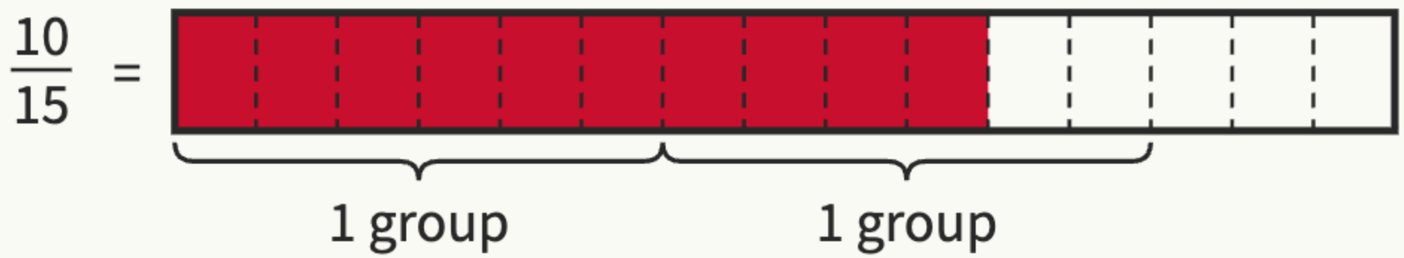
Next, we represent $\frac{10}{15}$ using the bar model.

$\frac{10}{15}$ =



Different Denominators

Use the quotative division model to evaluate $\frac{2}{3} \div \frac{2}{5}$.



Then we look for groups of size $\frac{6}{15}$, of which we can find 1 complete group in the shaded region.

For the second group, the shaded region only gives us $\frac{4}{6}$ of what is needed (we have 4 parts, but need 6).

So we have a total of $1\frac{4}{6}$ groups, which means that $\frac{2}{3} \div \frac{2}{5} = 1\frac{4}{6}$.

The Keep-Change-Flip Strategy

The strategies we have developed thus far work well, but having to draw the bar model to help us see the groups can be cumbersome and, in some cases, prohibitive. Thankfully, we have other strategies available.

Theorem 6.4.13 Keep-Change-Flip

Given two fractions $\frac{a}{b}$ and $\frac{c}{d}$, where $c \neq 0$,

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}.$$

Why Does Keep-Change-Flip Work?

In words, Theorem 6.4.13 essentially says that dividing by a fraction, $\frac{c}{d}$, is equivalent to multiplying by its **reciprocal**, $\frac{d}{c}$. Below, we examine why this is true.

By the definition of division, $\frac{a}{b} \div \frac{c}{d} = q$, where q is some rational number, if and only if $q \cdot \frac{c}{d} = \frac{a}{b}$.

Taking this and multiplying both sides by $\frac{d}{c}$ gives us the following.

$$q \cdot \frac{c}{d} = \frac{a}{b}$$

$$q \times \frac{c}{d} \times \frac{d}{c} = \frac{a}{b} \times \frac{d}{c}$$

$$q \times \frac{cd}{dc} = \frac{a}{b} \times \frac{d}{c}$$

$$q = \frac{a}{b} \times \frac{d}{c}$$

Using Keep-Change-Flip

Use Theorem 6.4.13 to evaluate $\frac{5}{6} \div \frac{2}{3}$.

$$\begin{aligned}\frac{5}{6} \div \frac{2}{3} &= \frac{5}{6} \times \frac{3}{2} \\ &= \frac{5 \times 3}{6 \times 2} \\ &= \frac{15}{12}\end{aligned}$$

Another Strategy for Dividing Fractions

Look at $\frac{6}{11} \div \frac{2}{11} = 3$. How could we get the 3 from the numerators?

The denominators are the same and the result can be obtained by simply dividing the numerators. In fact, we are also dividing the denominators, but we get 1.

Theorem 6.4.15

Given two fractions $\frac{a}{b}$ and $\frac{c}{d}$, where $c \neq 0$ and $d \neq 0$,

$$\frac{a}{b} \div \frac{c}{d} = \frac{a \div c}{b \div d}.$$

Dividing Straight Across

Theorem 6.4.15 essentially says that we can divide straight across in the same way that we multiply straight across. Its proof is in §6.4 of the notes.

Use Theorem 6.4.15 to evaluate $\frac{6}{35} \div \frac{2}{7}$.

$$\begin{aligned} \frac{6}{35} \div \frac{2}{7} &= \frac{6 \div 2}{35 \div 7} \\ &= \frac{3}{5} \end{aligned}$$

A Quotient of 2

Use the digits 1 to 9, each at most one time, to make two fractions whose quotient is 2. Use any strategy to show that your answer is correct. How many different pairs can your group find?

$$\frac{\square}{\square} \div \frac{\square}{\square} = 2$$

There are 16 pairs in all. Did another group find a pair that your group missed?

Find the Mistake

A student uses a bar model to find $\frac{3}{4} \div \frac{2}{5}$. The student rewrites the fractions as $\frac{15}{20}$ and $\frac{8}{20}$, finds 1 group of size $\frac{8}{20}$, and sees $\frac{7}{20}$ left. The student writes $\frac{3}{4} \div \frac{2}{5} = 1\frac{7}{20}$. What mistake did the student make? Find the correct answer, and use the bar model to show that it is correct.